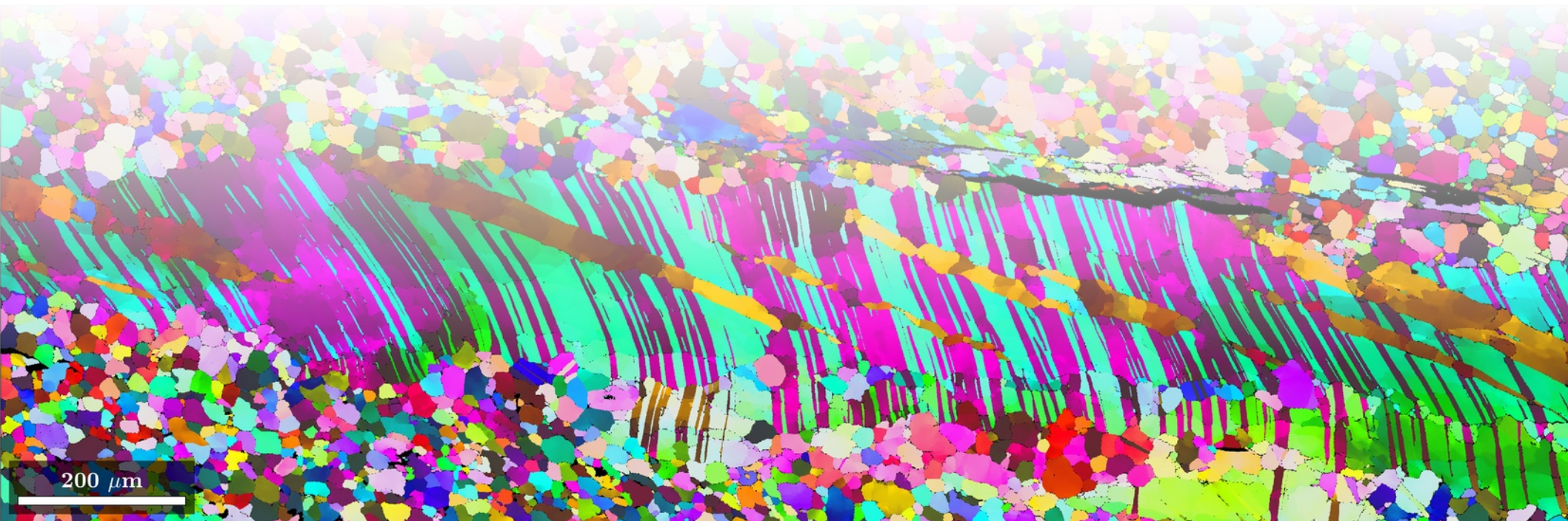


EBSD- reference frames and data import



1. EBSD/data import: formats

- importing EBSD data into mtex

- different formats supported
- non-proprietary and well documented formats preferred (consider during system acquisition)
- some formats are changed over time (e.g. OSC)
- generic interface can read almost arbitrary, human-readable data
- in any case, you may want to know the setup of your EBSD system!

mtex/interfaces/

loadEBSD_ACOM.m
loadEBSD_ang.m
loadEBSD_brukertxt.m
loadEBSD_crc.m
loadEBSD_csv.m
loadEBSD_ctf.m
loadEBSD_dream3d.m
loadEBSD_ebsdimage.m
loadEBSD_generic.m
loadEBSD_h5.m
loadEBSD_h5oia.m
loadEBSD_hdf5.m
loadEBSD_osc.m
loadEBSD_Oxfordcsv.m
loadEBSD_sor.m

1. EBSD/data import:

minimal example:

```
ebsd = EBSD.load('ice.ctf')
```

despite Mtex will guess the file format, many formats do not store important information in order to correctly interpret orientations

result:

Warning: .ctf files have usually inconsistent conventions for spatial coordinates and Euler angles. You may want to use one of the options 'convertSpatial2EulerReferenceFrame' or 'convertEuler2SpatialReferenceFrame' to correct for this

```
ebsd = EBSD
```

Phase	Orientations	Mineral	Color	Symmetry	Crystal reference frame
0	27273 (13%)	notIndexed			
1	184281 (87%)	Ice-Ih	LightSkyBlue	6/mmm	X a*, Y b, Z c*

Properties: bands, bc, bs, error, mad, x, y
Scan unit : um

1. EBSD/data import:

minimal example:

```
ebbsd = EBSD.load('ice.ctf', ...  
                  'convertEuler2SpatialReferenceFrame')
```

silence the warning! but
why and what does it
mean?

result:

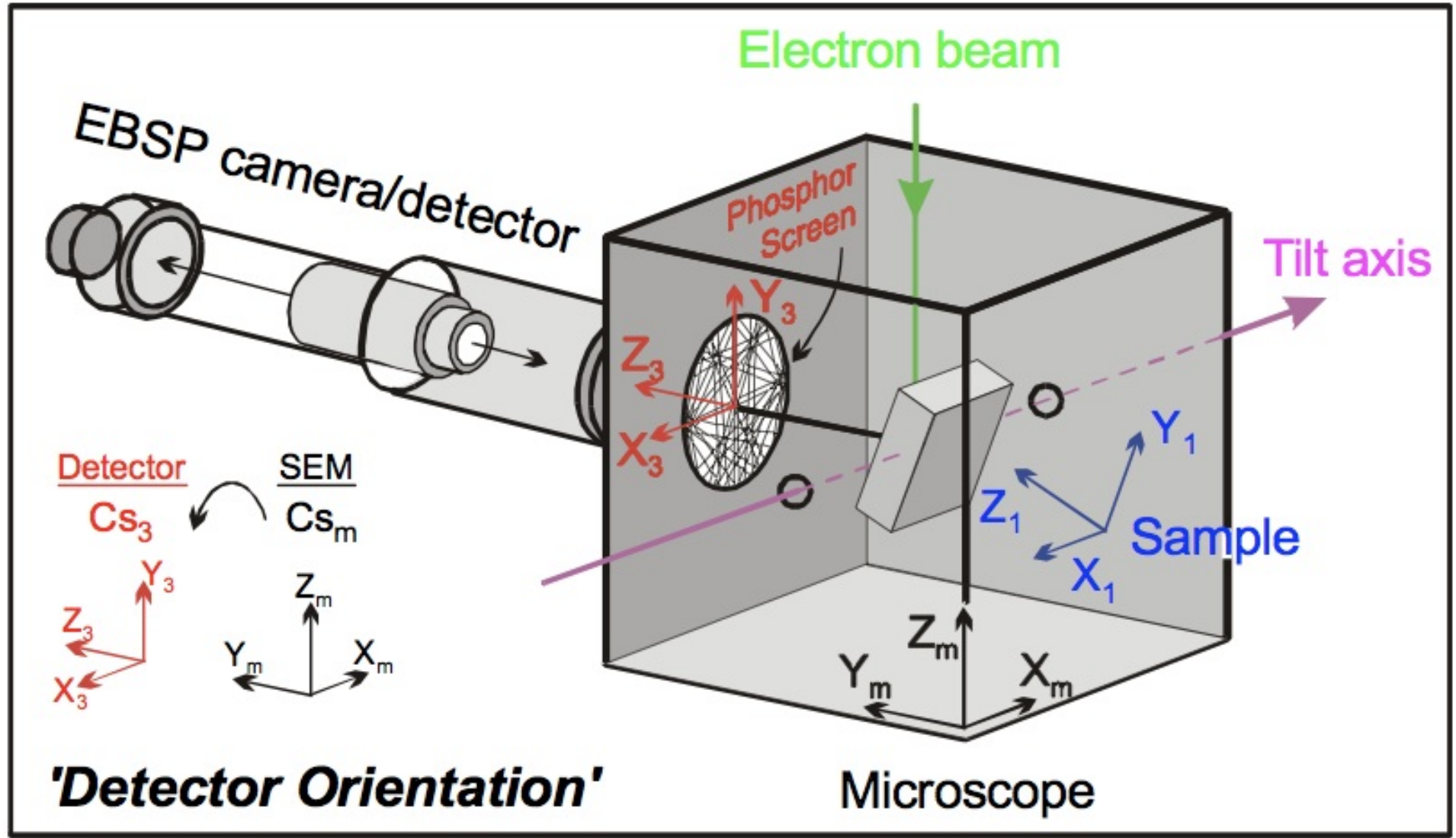
ebbsd = [EBSD](#)

Phase	Orientations	Mineral	Color	Symmetry	Crystal reference frame
0	27273 (13%)	notIndexed			
1	184281 (87%)	Ice-Ih	LightSkyBlue	6/mmm	X a*, Y b, Z c*

Properties: bands, bc, bs, error, mad, x, y
Scan unit : um

1. EBSD import: absolute orientations

Getting absolute orientations from EBSD:



1. EBSD import: absolute orientations

the problem:

in Mtex, there is only ONE x,y,z coordinate system, used to define pixel positions in a map, vector3d and rotation axes of Euler angles, ...

many vendors use at least TWO coordinate systems:

1) describing Euler angles

2) describing the map

and in most case $1 \neq 2$ (resulting even in inconsistent UI display of map/pf)

a simple attempt to correct this, either

transform the map coordinates:

```
... , 'convertSpatial2EulerReferenceFrame' , ...
```

or

transform the orientations:

```
... , 'convertEuler2SpatialReferenceFrame' , ...
```

Attention: the switches only assume some behavior found in many systems

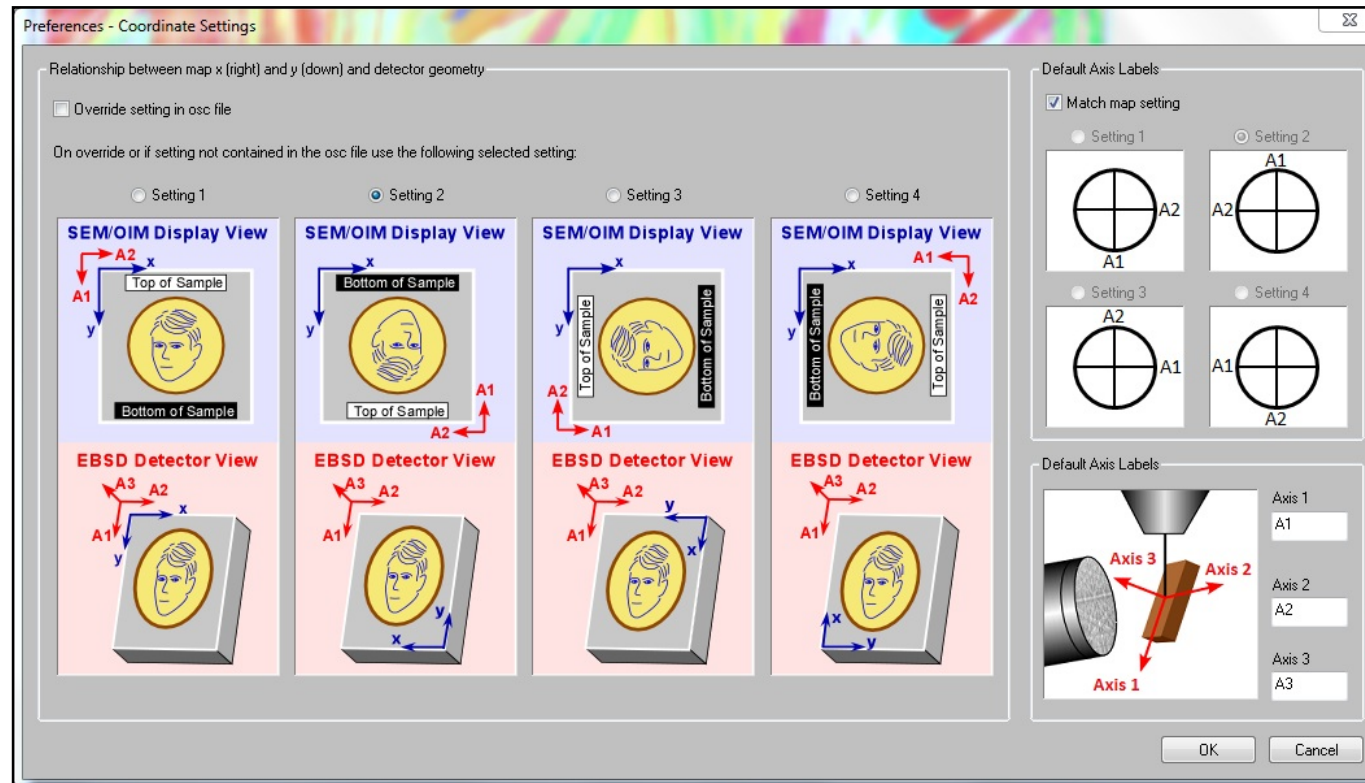
ctf/crc interface: 180° rotation around x-axis (! make sure this is correct at your system)

ang/osc: variable, default 180° around [110]-axis

1. EBSD import: absolute orientations

notes for EDAX/OIM systems:

- .ang files do not store the setting!
- .osc is binary and subject to change
- .hdf contains the setting

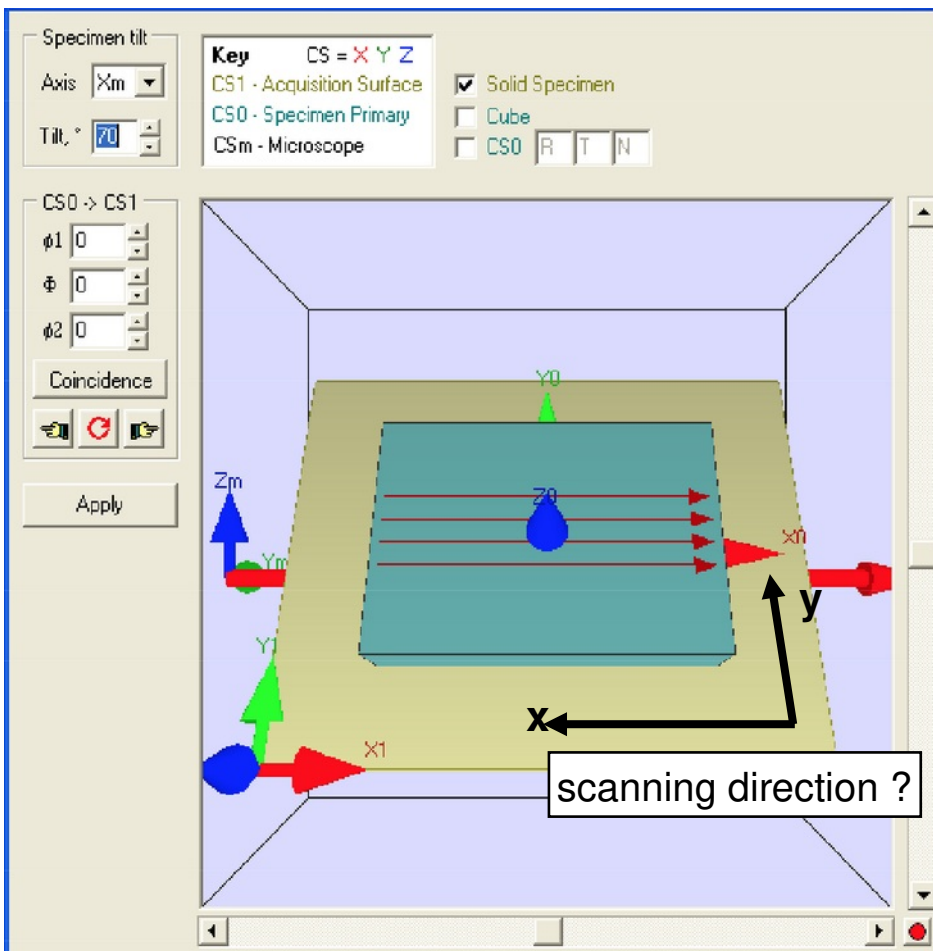


note: Mtex defaults to “setting 1” while “setting 2 ” seems to be most commonly used in OIM

EDAX/OIM

```
ebsd = loadEBSD_ang(fname,CS, ...  
    'convertEuler2SpatialReferenceFrame', 'setting 2')
```

1. EBSD import: absolute orientations



Channel5 / Oxford "virtual chamber" camera view?

historical Oxford systems:

- Euler angles: CS2 (crystal) -> CS1 (specimen)
- ctf stores angles wrt CS0 (good, if CS1->CS0 relation correct)
- old versions of Aztec (~2015) different export conventions did nothing on ctf (AcqE1,2,3 =0)

from .ctf header

```

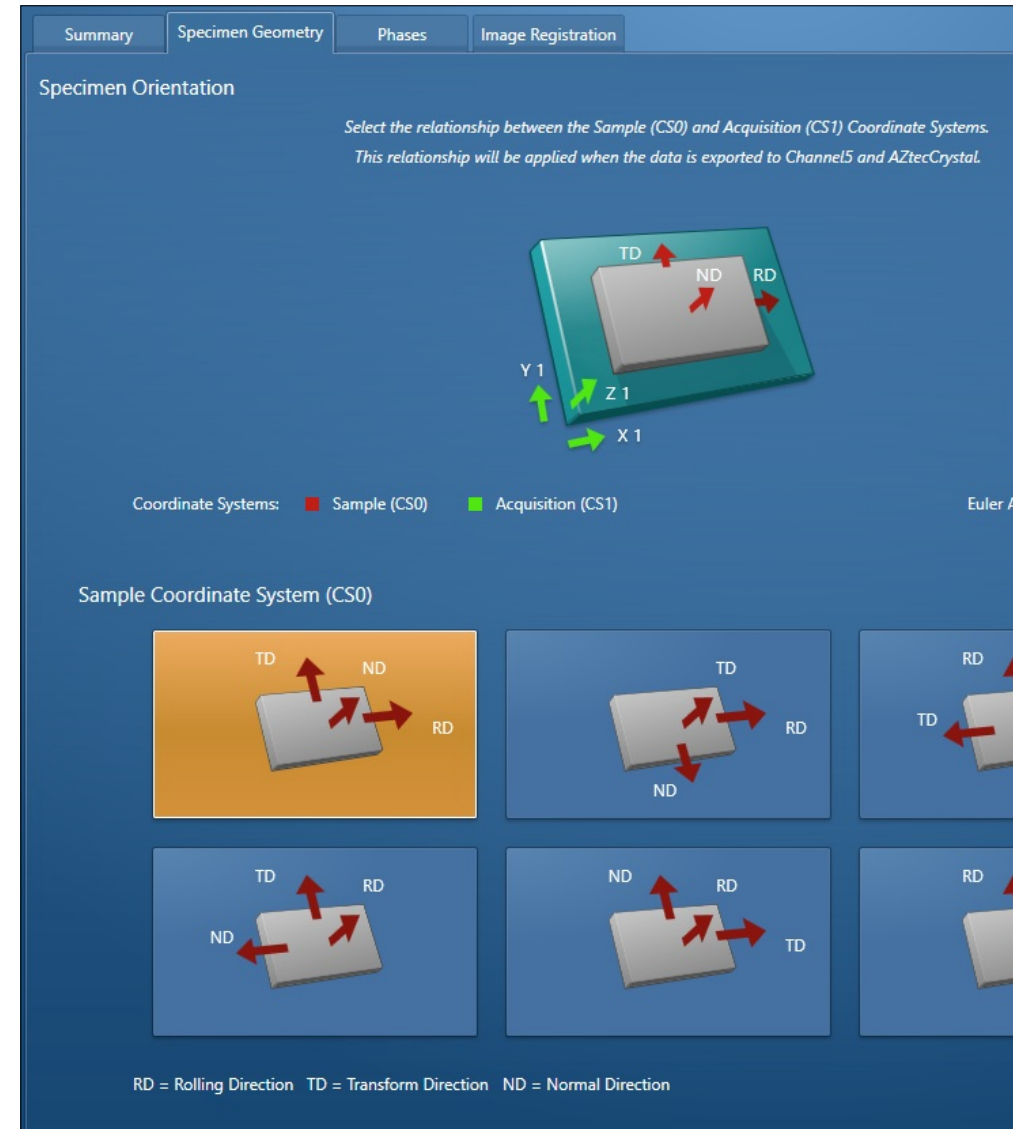
...
XCells    30
YCells    30
XStep      1
YStep      1
AcqE1      0
AcqE2      0
AcqE3      0
...

```


1. EBSD import: absolute orientations

notes for Oxford systems:

- .crc/.cpr format contains CS relation but binary and subject to change/deprecation
- .h5oia (hdf5) is well documented
<https://github.com/oinanoanalysis/h5oia> ,
contains all required information
- CS1-> CS0 can be set during export to h5oia
loasEBSD_h5oia should take care of this



see:

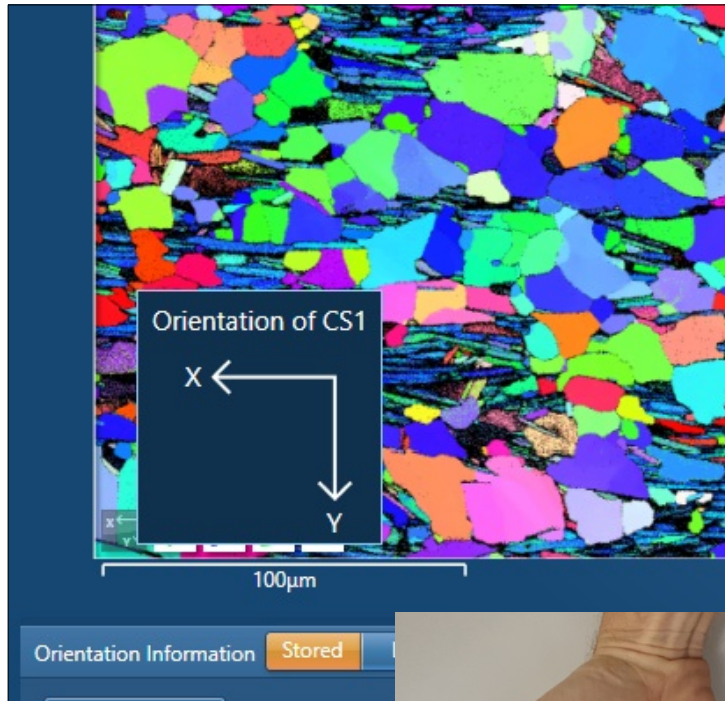
<https://github.com/oinanoanalysis/h5oia/blob/master/H5OINAFFile.md#-definition-of-coordinate-systems>

1. EBSD import: absolute orientations

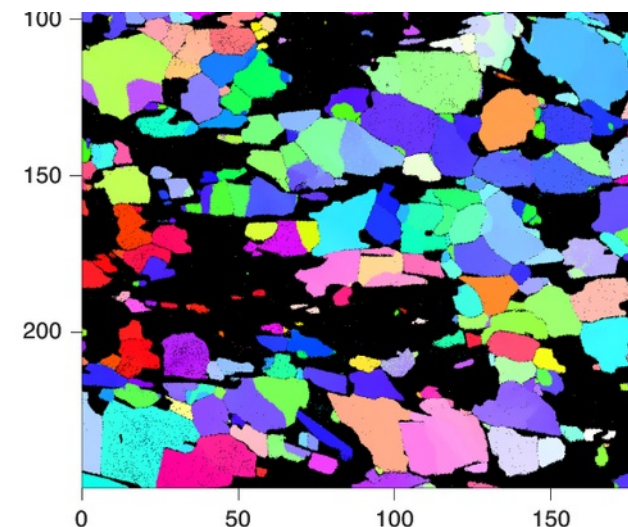
notes for Oxford systems:

it looks like this system needs the following correction:

```
ebbsd = loadEBSD_h5oina('data.h5oina')  
% in order to see the map as in the microscope  
plotx2east % set how2plot  
plotzIntoPlane  
% correct orientations  
rot = rotation.byAxisAngle(yvector,pi)  
ebbsd=rotate(ebbsd,rot,'keepXY')
```



mtex



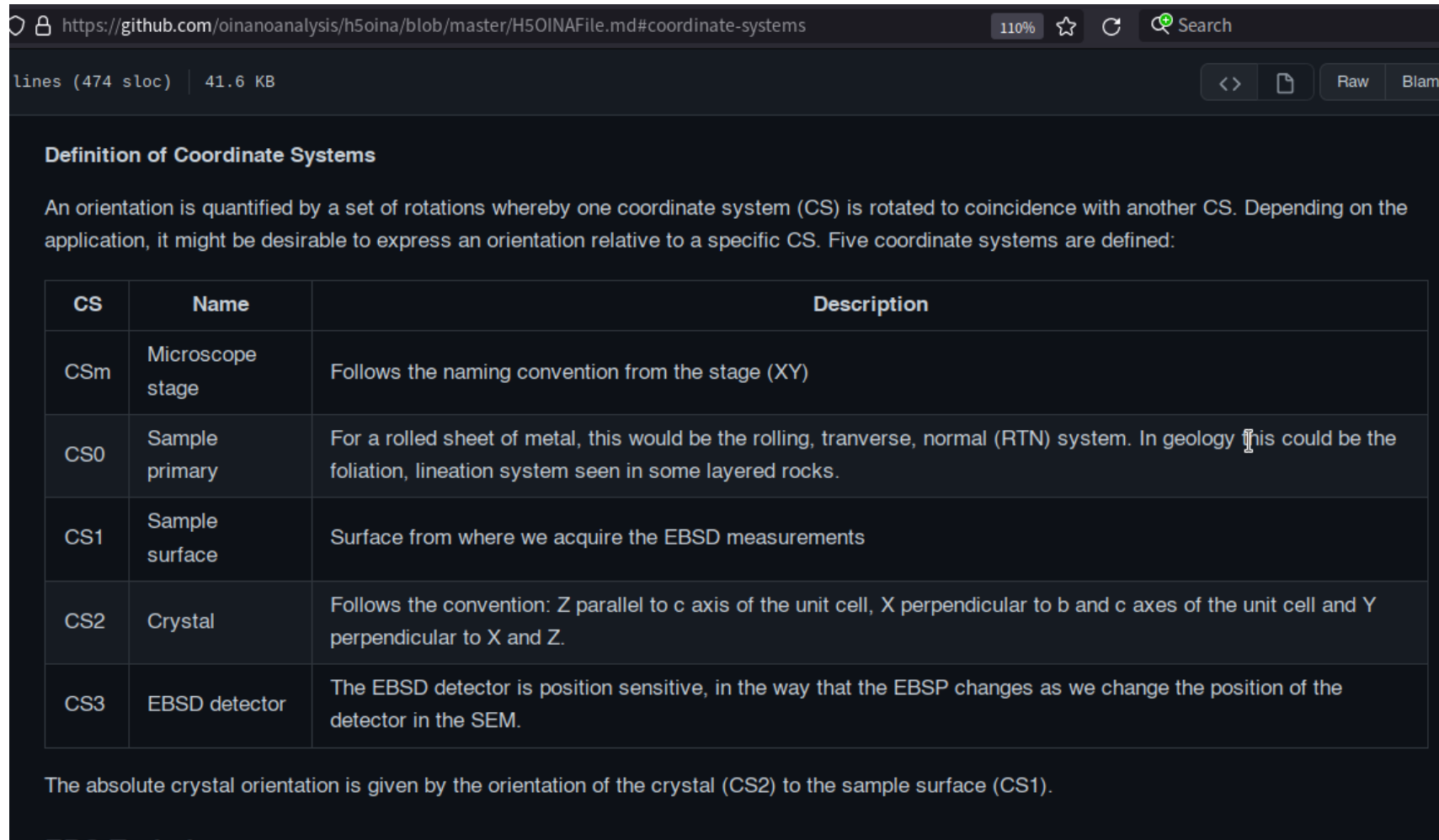
see:

<https://github.com/oianoanalysis/h5oina/blob/master/H5OINAFFile.md#-definition-of-coordinate-systems>

1. EBSD import: absolute orientations

notes for Oxford systems:

- .h5oina (hdf5) is well documented
<https://github.com/oinanoanalysis/h5oina> ,
contains all required information
- supported by loadEBSD_h5oina.m



lines (474 sloc) | 41.6 KB

Definition of Coordinate Systems

An orientation is quantified by a set of rotations whereby one coordinate system (CS) is rotated to coincidence with another CS. Depending on the application, it might be desirable to express an orientation relative to a specific CS. Five coordinate systems are defined:

CS	Name	Description
CSm	Microscope stage	Follows the naming convention from the stage (XY)
CS0	Sample primary	For a rolled sheet of metal, this would be the rolling, transverse, normal (RTN) system. In geology this could be the foliation, lineation system seen in some layered rocks.
CS1	Sample surface	Surface from where we acquire the EBSD measurements
CS2	Crystal	Follows the convention: Z parallel to c axis of the unit cell, X perpendicular to b and c axes of the unit cell and Y perpendicular to X and Z.
CS3	EBSD detector	The EBSD detector is position sensitive, in the way that the EBSD changes as we change the position of the detector in the SEM.

The absolute crystal orientation is given by the orientation of the crystal (CS2) to the sample surface (CS1).

1. EBSD import: absolute orientations

Getting absolute orientations from EBSD:

<https://groups.google.com/d/msg/mtexmail/EpOQo04KsmM/ugvOYTOqBgAJ>

~ 2014/15: 7 of 8 tested labs (from one manufacturer) produced 180° rotated orientation, (problems known since 2009 or earlier)

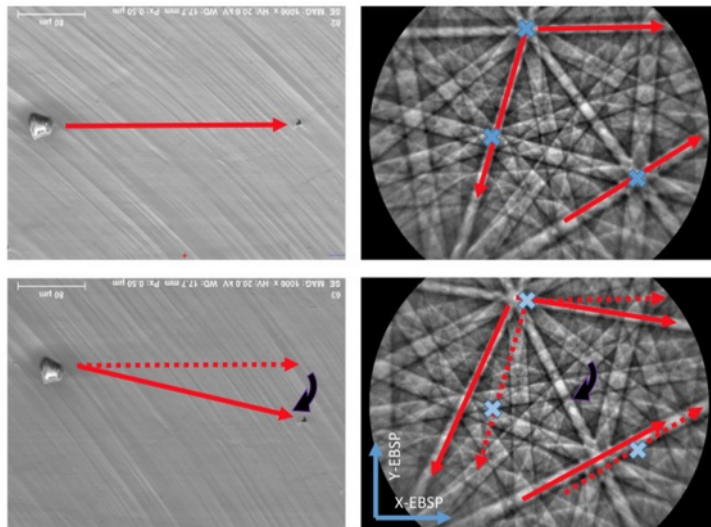
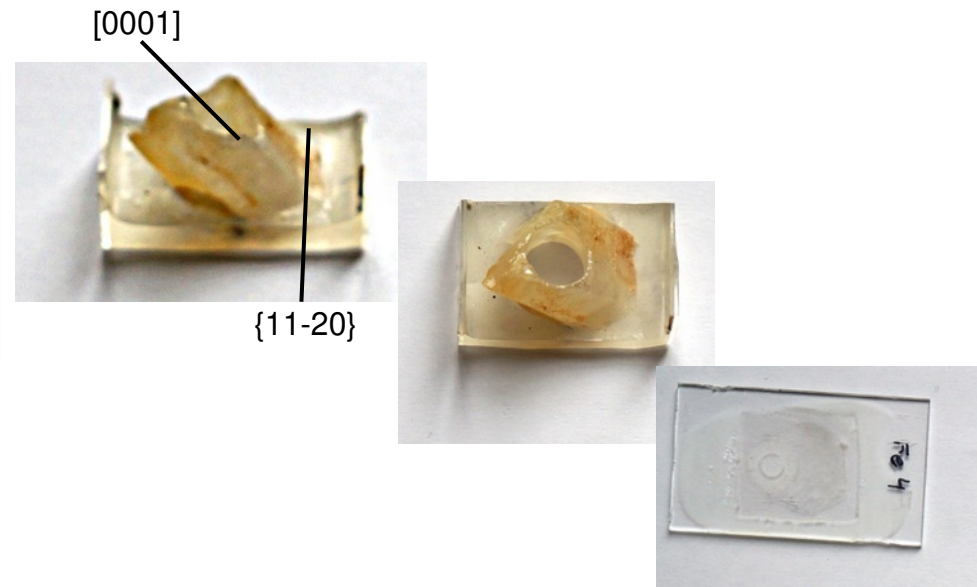
2021: many vendors take care of this (at least it's mentioned in manuals)

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Absolute orientations from EBSD measurements - as easy as it seems?

Rüdiger Kilian (1), Michel Bestmann (2), Renée Heilbronner (1,3)





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Tutorial: Crystal orientations and EBSD — Or which way is up?

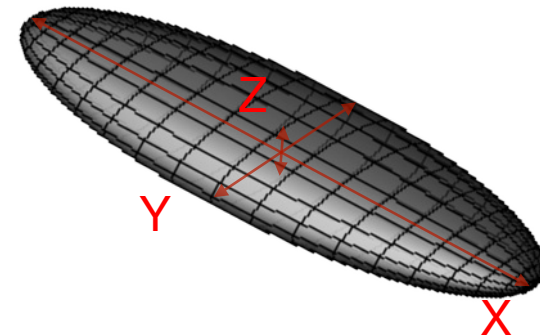
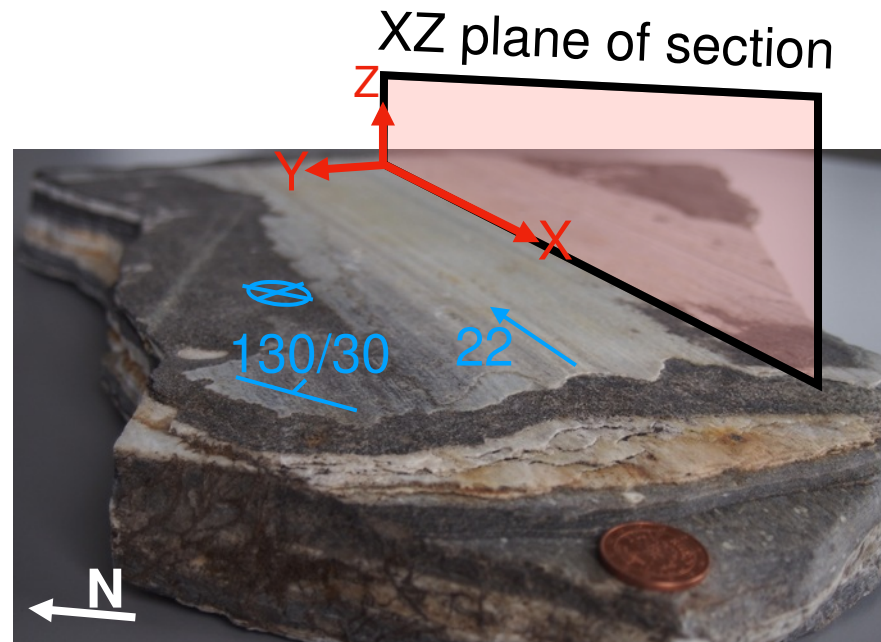
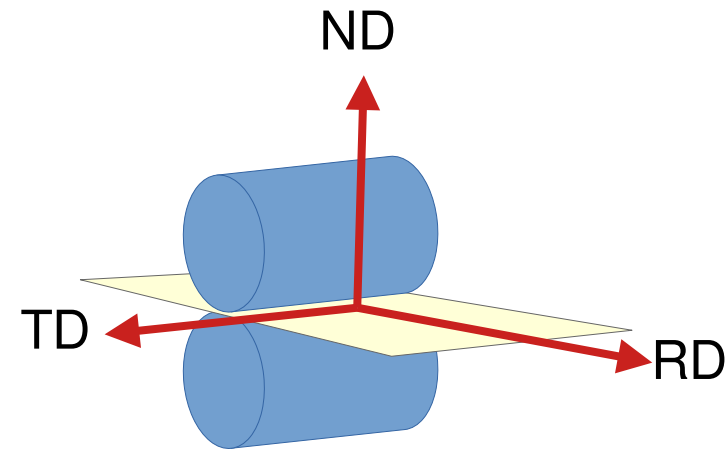
T.B. Britton ^{a,*}, J. Jiang ^a, Y. Guo ^{b,1}, A. Vilalta-Clemente ^b, D. Wallis ^c, L.N. Hansen ^c,
A. Winkelmann ^d, A.J. Wilkinson ^b



1. EBSD: plotting conventions

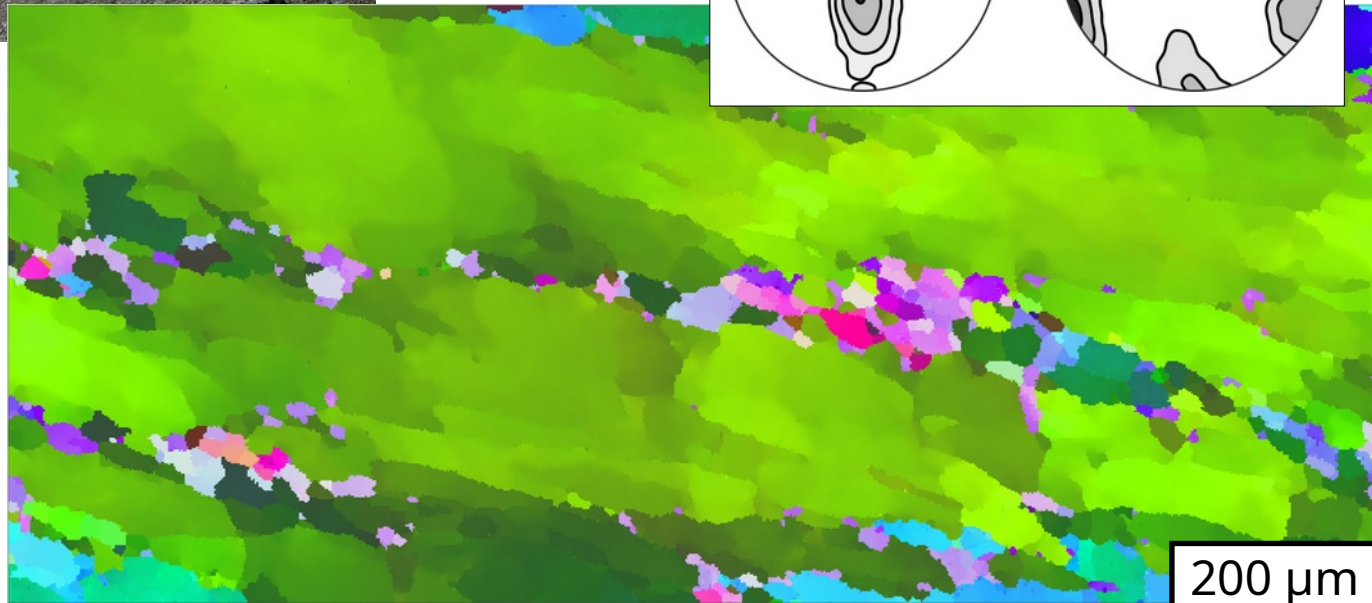
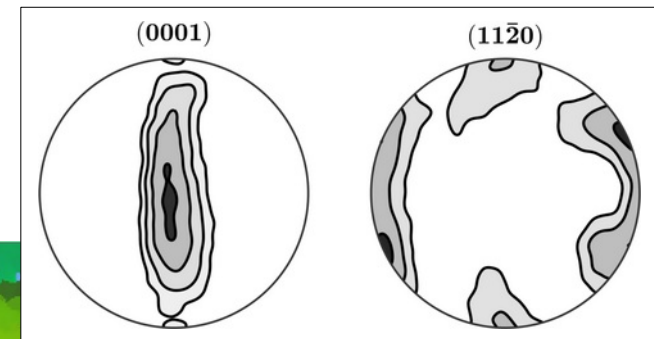
Why might we want to specify the plotting convention?

- we might like our EBSD map / pf to appear in a specific way on the screen
- plotting convention sets relation of the x,y,z coordinate system used by Mtex wrt the screen
- Mtex does not know (yet) about TD, RD, ND or X, Y, Z reference frames used in metal processing or structural geology!



1. EBSD/data import: reference frames

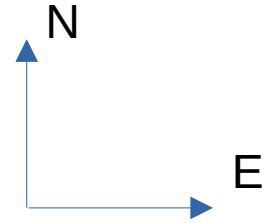
Getting absolute orientations from EBSD: Why it might matter (for geologists)



200 μm

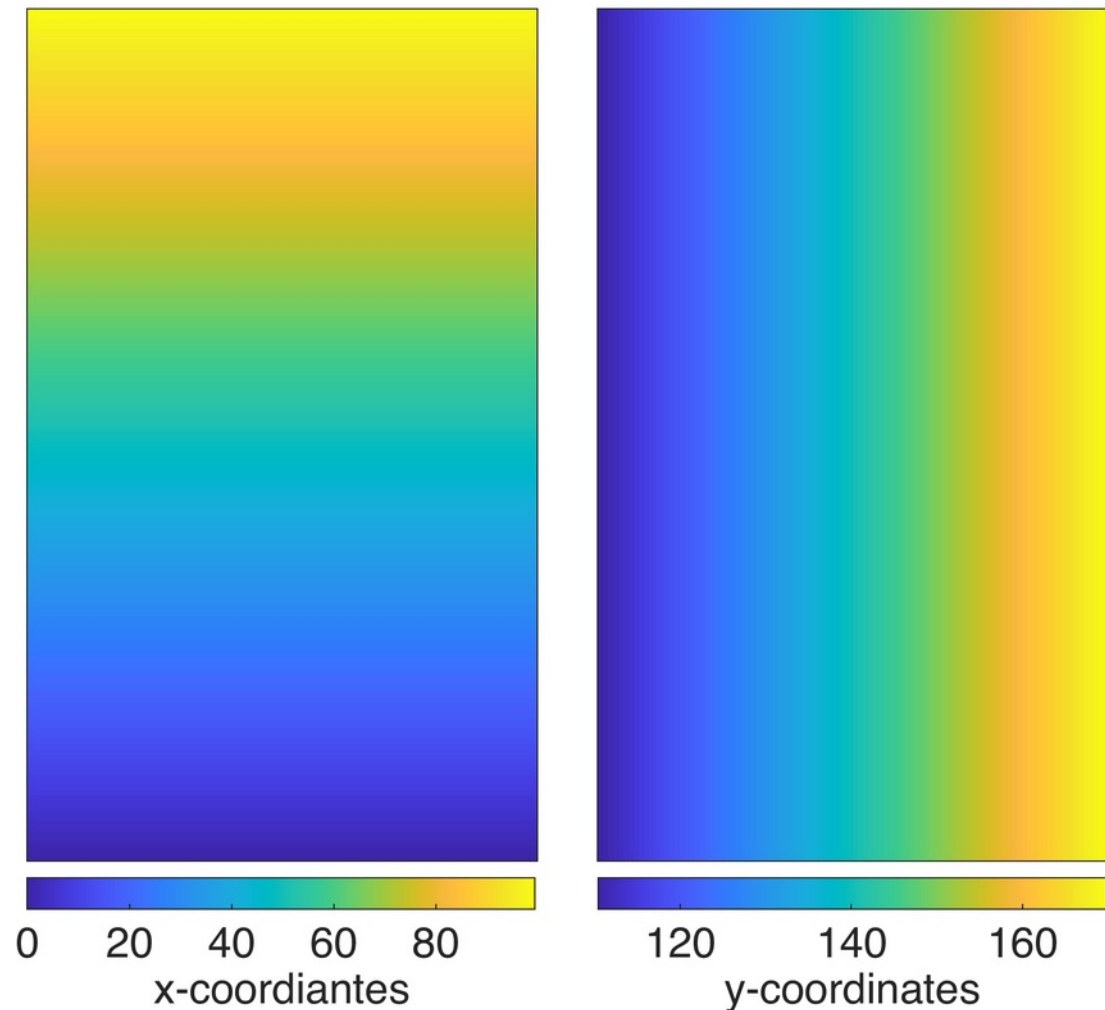
1. EBSD/data import: reference frames

Setting plotting convention: does not change x,y coordinates of the EBSD object.



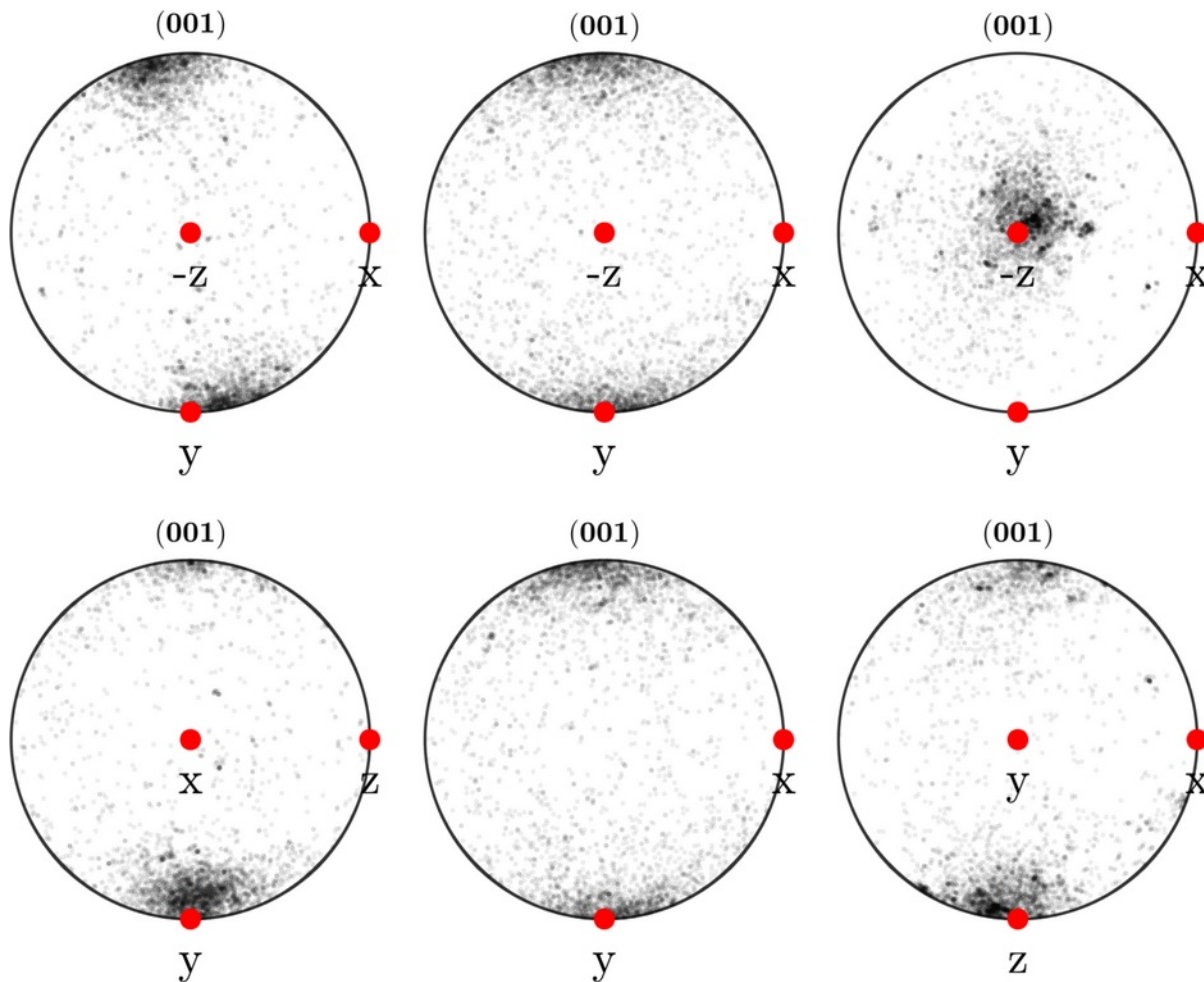
X: north - Z: intoPlane

```
plotx2north  
plotzIntoPlane
```



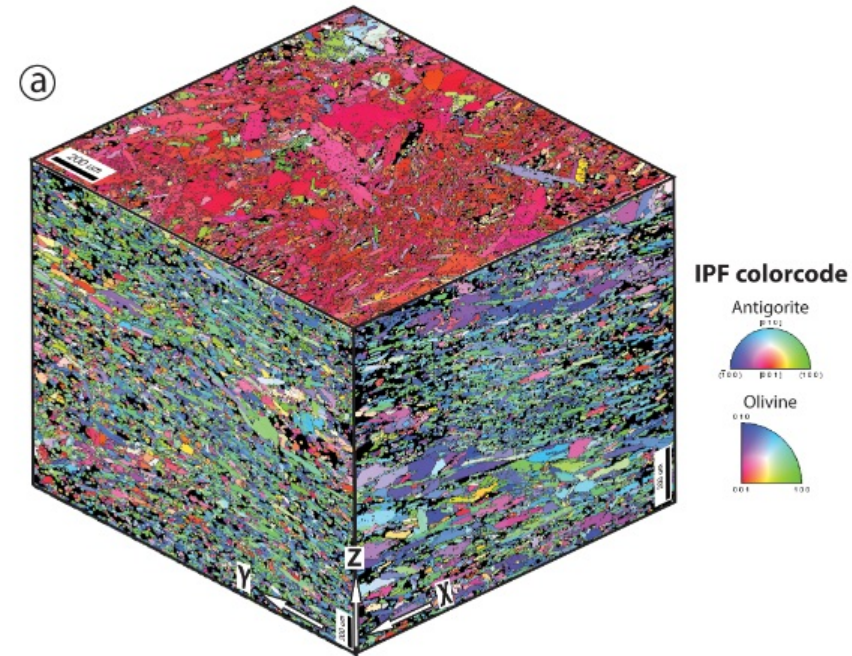
1. EBSD/data import: reference frames

pole figure relative to the maps



pole figure relative to the "world"

Orientation map



Morales et al, 2018